Premature Accountability Convergence:
AI-Mediated Responsibility in Sociotechnical Organizations

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Abstract

Responsibility in regulated organizations does not emerge only after failure; it is continuously produced through anticipatory alignment, procedural embedding, and retrospective stabilization. As artificial intelligence (AI) systems become integrated into documentation and evaluative infrastructures, responsibility is increasingly mediated through computational representation. Yet AI does not introduce new accountability structures. Instead, it reshapes the temporal dynamics through which responsibility becomes stabilized.

This paper introduces Premature Accountability Convergence (PAC), a conditional acceleration dynamic within the responsibility lifecycle. PAC occurs when computational mediation compresses interpretive space such that responsibility stabilizes before causal ambiguity has been meaningfully examined. Under conditions of representational compression, epistemic authority amplification, and temporal contraction of sensemaking, responsibility may converge while plausible alternative explanations remain insufficiently explored.

By conceptualizing AI as a representational accelerator rather than an autonomous decision-maker, this framework shifts attention from algorithmic bias toward the temporal architecture of accountability. PAC extends sociotechnical theories of responsibility by specifying how computational infrastructures intensify stabilization dynamics already embedded within organizational systems. Rather than determining who is responsible, AI may reshape how quickly and how narrowly responsibility becomes administratively actionable.

Introduction

Artificial intelligence (AI) systems are increasingly embedded within organizational

documentation, evaluation, and compliance infrastructures. These systems generate structured representations such as summaries, classifications, anomaly flags, and predictive indicators that enter directly into accountability workflows. While often described as decision-support tools, AI systems operate at a deeper infrastructural level. They intervene in the representational architecture through which responsibility becomes legible, assessable, and administratively actionable. Scholars of algorithmic governance have observed that computational systems increasingly structure how information is classified, evaluated, and rendered actionable within organizational settings (Kitchin, 2017; Burrell, 2016).

Responsibility in regulated organizations is not a singular event but a temporally structured accomplishment produced through ongoing organizational sensemaking processes (Weick, 1995; Maitlis & Christianson, 2014). It is anticipatorily aligned before action, procedurally embedded during execution, and retrospectively stabilized under conditions of ambiguity and closure pressure. Prior work has examined anticipatory structuring of responsibility before action and retrospective attribution through investigative reasoning after events occur.

What remains theoretically underspecified is the temporal dimension of convergence itself. Although prior research explains why responsibility converges, considerably less attention has been given to when convergence occurs relative to the persistence of causal ambiguity. As AI systems become embedded within documentation and evaluative processes, interpretive space may narrow earlier in the sensemaking sequence. Convergence may occur more rapidly not because actors intend closure or institutions become more rigid but because computational mediation compresses representational diversity and accelerates inscription.

Existing scholarship on algorithmic accountability has primarily examined opacity, which refers to the difficulty of understanding how computational systems generate outputs (Pasquale, 2015; Burrell, 2016). It has also examined the amplification or reproduction of social biases within training data and classification schemes (Ananny & Crawford, 2018; Kitchin, 2017). A related line of inquiry has examined how AI systems displace responsibility onto human operators who retain nominal oversight but exercise diminished causal control. Elish (2019) refers to this dynamic as “moral crumple zones.” More recent work has extended this concern to the organizational infrastructures surrounding AI deployment, including the documentation practices and audit frameworks through which accountability is enacted (Raji et al., 2020).

PAC addresses a distinct but complementary question. Regardless of whether computational outputs are accurate or transparent, and regardless of where responsibility is ultimately assigned, how does their integration into organizational documentation and evaluative infrastructures reshape the timing and scope of responsibility stabilization? The issue is not primarily whether AI produces correct attributions or generates crumple zones for human operators. Rather, the question is whether the tempo and breadth of convergence allow sufficient deliberation before stabilization becomes institutionally binding.

This paper introduces Premature Accountability Convergence (PAC) to conceptualize this acceleration mechanism. PAC identifies the conditions under which computational mediation compresses interpretive space and accelerates responsibility stabilization within the responsibility lifecycle. In doing so, the framework advances a temporal perspective on organizational accountability.

Responsibility as a Multi-Stage Organizational Lifecycle

Responsibility in regulated sociotechnical organizations unfolds across temporally distinct yet structurally interconnected stages of organizational action and evaluation. Rather than emerging only at the moment of investigation or formal evaluation, responsibility is produced progressively through anticipatory alignment, procedural embedding, and retrospective stabilization. These stages form a lifecycle through which accountability expectations are generated, operationalized, and ultimately consolidated into administratively actionable forms.

Anticipatory Structuring of Responsibility

In regulated environments, organizational actors rarely act without anticipating future evaluation. Accountability expectations therefore operate prospectively, shaping how obligations, risks, and acceptable conduct are interpreted before action occurs. Anticipatory alignment emerges when actors orient their behavior toward expected evaluative standards, interpreting instructions, exercising discretion, and assessing risk with awareness that their actions may later be scrutinized. At this stage, convergence occurs prospectively as interpretive variance narrows through alignment with anticipated evaluative criteria rather than through formal attribution.

Procedural Embedding During Execution

During execution, responsibility expectations become embedded within documentation systems, reporting architectures, and procedural infrastructures. Standard operating procedures, logging requirements, validation protocols, and compliance frameworks structure how events are recorded and translated into organizational memory within broader audit and verification infrastructures (Power, 1997). Procedural embedding stabilizes responsibility not by assigning blame but by structuring the evidentiary pathways through which later attribution will occur. Documentation formats define what counts as relevant information, how deviations are categorized, and which explanatory variables become visible within formal records. Convergence at this stage remains incomplete but cumulative as interpretive possibilities are channeled into administratively legible categories.

Retrospective Stabilization of Responsibility

After outcomes occur, responsibility becomes articulated and consolidated through investigations, reports, evaluative documentation, and corrective action processes. Under conditions of causal ambiguity, organizations must translate distributed events into administratively actionable accounts. Investigative reasoning often unfolds under closure pressure, as ambiguous causal pathways must be rendered determinate in order to justify corrective action, compliance responses, or performance evaluation. Through this process, interpretive alternatives are gradually reduced as particular explanatory narratives acquire institutional authority. Retrospective stabilization therefore marks the formal consolidation of responsibility convergence, as ambiguity becomes condensed into administratively durable accounts embedded within documentation systems and organizational memory.

Understanding this lifecycle clarifies how accountability becomes temporally organized prior to the introduction of computational mediation.

Computational Mediation within the Responsibility Lifecycle

AI systems are increasingly embedded within procedural workflows, documentation regimes, and evaluative infrastructures across regulated domains. Rather than functioning as autonomous decision-makers, these systems operate primarily as representational mediators that translate organizational events into structured, recordable, and evaluable forms. Computational outputs such as summaries, classifications, anomaly flags, pattern detections, and predictive

indicators do not independently assign responsibility. However, they shape how events become legible within organizational documentation and evaluative processes.

Importantly, computational mediation does not introduce responsibility structures ex nihilo. The anticipatory alignment, procedural embedding, and retrospective stabilization of responsibility precede the integration of AI systems. Organizational infrastructures already generate expectations of accountability and stabilize attribution through investigative processes. What computational mediation alters is the tempo through which convergence unfolds. Algorithmically generated representations often carry the perceived authority of formalized calculation, which can narrow interpretive divergence earlier in the sensemaking process. Certain causal pathways may therefore be foregrounded while others remain backgrounded, shaping which interpretations become administratively actionable. As a result, responsibility expectations may align and stabilize more rapidly when mediated through computationally structured representations.

Premature Accountability Convergence (PAC)

PAC identifies a structural acceleration dynamic that becomes visible under conditions of computational mediation. PAC occurs when representational compression, perceived epistemic authority, and temporal contraction interact to reduce interpretive variance before causal ambiguity has been meaningfully examined. The responsibility lifecycle itself remains intact. What shifts is the timing and scope through which responsibility stabilization occurs.

Within this framework, the term “premature” carries a specific evidentiary meaning. Stabilization is premature when it forecloses causal alternatives that remain evidentially plausible. In other words, responsibility converges before the evidentiary record has been sufficiently interrogated to exclude distributed, systemic, or multi-causal explanations. This claim is not simply about speed. Rapid stabilization is unremarkable when the evidentiary record has been genuinely exhausted and interpretive plurality has been meaningfully examined. Prematurity instead refers to an evidential and procedural condition.

More specifically, convergence is premature when at least one of the following conditions holds:

(a) alternative causal accounts remain plausible given the available evidence;

(b) (the documentary record was structured in ways that systematically rendered certain

explanations invisible prior to investigation;

(c) the deliberative interval was compressed before distributed or systemic explanations could acquire institutional salience.

These conditions are analytically distinguishable from one another and from ordinary investigative closure. This distinction provides the basis for the framework's empirical tractability.

Speed and breadth represent analytically distinct dimensions of this dynamic. Representational compression constrains breadth by narrowing the range of visible explanatory pathways. Temporal contraction reduces the deliberative interval available for exploring those pathways. Epistemic authority amplification operates across both dimensions by increasing the institutional weight granted to compressed and accelerated outputs. PAC therefore specifies the conditions under which narrowing scope and accelerated closure combine to produce premature convergence within the responsibility lifecycle.

PAC does not claim that AI inherently distorts accountability. Rather, it specifies a conditional infrastructural dynamic. When structured computational outputs enter documentation and evaluative systems early in the sensemaking sequence, responsibility may converge more rapidly and with diminished interpretive plurality. This framing is complementary to but distinct from Elish's (2019) concept of moral crumple zones, which describes the misattribution of responsibility to human operators who retain nominal oversight but limited causal control. Whereas the crumple zone framework focuses on the destination of responsibility, meaning who ultimately absorbs it, PAC focuses on the architecture of convergence, meaning how quickly and how narrowly stabilization occurs. Both dynamics may operate simultaneously, but they remain analytically separable and therefore open to different forms of intervention.

Representational Compression

Computational systems translate complex and distributed organizational events into structured outputs such as summaries, classifications, risk scores, anomaly flags, and predictive indicators. In doing so, they shape which aspects of organizational events become visible and relevant within documentation systems (Gillespie, 2014). This process compresses interpretive multiplicity into standardized representational forms. It is structurally analogous to what Espeland and Stevens (1998) describe as commensuration, the translation of qualitatively distinct

phenomena into a common metric. A consequential effect of commensuration is that non-quantifiable or categorically ambiguous aspects of events may become institutionally invisible. Bowker and Star (1999) similarly demonstrate that classification infrastructures do not merely record organizational reality but actively shape what becomes visible, categorizable, and actionable within institutional processes. This dynamic becomes particularly consequential when classification is performed computationally and at scale.

Representational compression enhances administrative clarity and traceability. However, it also pre-selects which explanatory variables remain visible within formal documentation. As events are formatted into predefined categories, the range of plausible causal accounts may narrow prior to full deliberation. Interpretive variance does not disappear; rather, it becomes institutionally backgrounded. Representational compression therefore operates as a primary structural driver of PAC. By formatting events into administratively legible containers before interpretive deliberation has fully unfolded, it constrains the breadth of interpretive alternatives that remain institutionally visible during subsequent sensemaking and investigative processes.

Epistemic Authority Amplification

Algorithmically generated outputs often carry perceived legitimacy derived from formalized calculation, technical infrastructure, and data-driven framing (Porter, 1995). When computational representations are treated as epistemically authoritative, interpretive contestation may diminish as actors treat those representations as evidentiary anchors. Research on algorithmic accountability further suggests that computational systems frequently acquire institutional authority because their outputs appear procedurally objective and technically neutral within documentation and governance processes (Ananny & Crawford, 2018; Raji et al., 2020).

Epistemic authority amplification is analytically distinct from representational compression. Compression operates on the content of representation by narrowing the range of visible explanatory pathways. Authority amplification operates on the institutional status of those representations by increasing the weight granted to them within formal accountability processes. A system may generate compressed representations without those outputs being granted epistemic authority, for example when computational outputs are explicitly flagged as preliminary or subject to human review. Conversely, authority amplification may occur even

when compression is moderate if organizational actors treat structured outputs as definitive anchors for attribution without sustained interpretive contestation.

Temporal Contraction of Sensemaking

Computational mediation frequently reduces the interval between event occurrence and formal inscription. Automated logging systems, real-time dashboards, and immediate classification tools shorten the deliberative window within which alternative interpretations may circulate before formal documentation occurs. When documentation is generated rapidly and embedded early within evaluative infrastructures, preliminary interpretations may acquire institutional durability before distributed explanations are fully articulated.

Temporal contraction is analytically distinct from both representational compression and epistemic authority amplification. Compression constrains which interpretations are visible. Authority amplification constrains how readily those interpretations are contested. Temporal contraction constrains when stabilization occurs relative to the interpretive process. A highly compressed representation may be generated slowly, with extended deliberation, without triggering PAC. Conversely, rapid inscription may occur without categorical compression if documentation preserves narrative plurality. Temporal contraction therefore amplifies the acceleration dynamic initiated by representational compression by ensuring that early compressed outputs become embedded within evaluative infrastructures before alternative accounts have had opportunity to develop institutional salience.

The Interaction of PAC Mechanisms

These three mechanisms are analytically distinct, although they frequently interact in practice. Representational compression can occur without triggering PAC if computational outputs are treated as provisional rather than authoritative. Temporal contraction may likewise occur without representational compression, since rapid documentation can shorten the deliberative window without imposing categorical formatting on interpretive content. Epistemic authority amplification may intensify convergence dynamics even when outputs are generated slowly if structured representations are subsequently treated as definitive anchors for attribution.

PAC therefore requires not merely one of these conditions but their joint activation. Compression narrows the interpretive field, authority amplification anchors explanatory frames,

and temporal contraction forecloses the opportunity for alternative interpretations to develop institutional salience. When any one of these conditions is attenuated, PAC intensity is expected to diminish. This may occur when computational outputs are explicitly framed as preliminary, when deliberative review periods are institutionally protected, or when documentation systems preserve alternative causal accounts alongside primary classifications.

This configurational structure also specifies the conditions under which PAC is falsifiable. Convergence acceleration should not be observed in settings where computational outputs are treated as non-authoritative, where deliberative intervals are institutionally protected, or where representational systems explicitly preserve multiple causal accounts.

Illustrative Case: Automated Deviation Logging and Accountability Stabilization in Pharmaceutical Manufacturing

The following case is offered as a theoretical illustration rather than as empirical evidence for the PAC framework. Its purpose is to demonstrate how the three mechanisms may interact within a regulated environment where computational documentation is directly integrated into post-event investigative workflows. No claims are made regarding the frequency or generalizability of the described patterns. Those questions require empirical investigation.

Pharmaceutical manufacturing provides a particularly clear setting for illustrating PAC because all three mechanisms are structurally present within formalized post-event investigation processes. Facilities regulated by agencies such as the U.S. Food and Drug Administration (FDA) and the European Medicines Agency (EMA) are required to document, investigate, and resolve manufacturing deviations through structured Corrective and Preventive Action (CAPA) systems. These systems increasingly incorporate computational tools such as automated batch record review, electronic deviation logging, statistical process control alerts, and AI-assisted trend detection. These tools generate structured outputs that enter directly into investigation workflows (FDA, 2006).

Representational compression occurs when electronic batch records and automated monitoring systems translate complex manufacturing sequences involving equipment conditions, material variability, environmental factors, and operator judgment into structured deviation categories and alert flags. These standardized formats define which variables are formally captured and which remain outside the documented record. Systemic factors such as training system

adequacy, staffing levels, procedural ambiguity, or equipment lifecycle conditions may not appear as discrete fields within deviation management software even when they materially contribute to an event. The evidentiary field available to investigators therefore becomes pre-structured around variables that automated systems are designed to capture. These variables often foreground operator actions and discrete process parameters rather than distributed organizational conditions (Bowker & Star, 1999). This dynamic reflects the commensuration process described by Espeland and Stevens (1998), in which complex phenomena are translated into administratively tractable categories that inevitably exclude what cannot be measured within the scheme.

Epistemic authority amplification emerges as electronically generated records including timestamped system alerts, automated out-of-specification flags, and equipment log entries acquire institutional weight within CAPA documentation. Regulatory expectations that investigations be evidence-based and traceable create pressure to anchor causal accounts in documentary evidence rather than contextual inference. Computational records, by virtue of their apparent precision and automatic generation, often satisfy these evidentiary expectations more readily than qualitative accounts of systemic conditions. Investigators operating under regulatory scrutiny may therefore treat automated records as the primary evidentiary layer, concentrating attribution on actor-level actions that are most directly traceable within those records.

Temporal contraction occurs when automated systems generate formal deviation records immediately after an event occurs, often before informal interpretive discussions among production staff, quality personnel, and process engineers have fully developed. In facilities with real-time monitoring infrastructure, the formal record may be generated within minutes of a deviation trigger, initiating investigation timelines under regulatory clock constraints. This rapid inscription, combined with regulatory deadlines for CAPA completion, compresses the deliberative interval within which alternative causal explanations might develop and be formally examined.

This illustrative scenario is consistent with patterns occasionally noted in pharmaceutical quality literature, where CAPA investigations are reported to frequently identify immediate causes such as operator error, procedural deviation, or equipment failure rather than broader systemic conditions (FDA, 2006). Whether such patterns reflect PAC dynamics rather than investigator intent, cognitive bias, or genuine causal structures remains an open empirical question. The PAC framework provides a structural explanation for why such patterns might arise systematically.

However, evaluating this claim requires empirical analysis of actual investigation records across regulated environments.

Theoretical Propositions: Premature Accountability Convergence

Proposition 1. Representational Compression

The greater the degree of computational representational compression within organizational documentation systems, the more likely interpretive space will narrow prior to full causal exploration.

Representational compression refers to the extent to which complex, distributed, and ambiguous events are translated into standardized, categorical, or summarized formats. As organizational documentation systems increasingly rely on structured computational outputs such as classification schemas, risk categories, and summary dashboards, interpretive possibilities become organized within predefined representational structures. This structuring enhances administrative clarity and efficiency but can simultaneously restrict the range of causal alternatives that remain visible within formal documentation. When ambiguity is translated into categorical outputs, a process Espeland and Stevens (1998) describe as inherently exclusionary, alternative explanatory pathways may lose institutional salience before they are fully examined.

Empirically, this proposition could be evaluated through comparative analysis of post-event investigation reports from organizations with varying levels of computational documentation density. Representational compression may be operationalized by coding the proportion of evidentiary content derived from structured computational outputs relative to unstructured qualitative materials such as staff testimony, observational notes, and contextual narrative accounts. A positive association between computational documentation density and concentration of individual-level attribution would provide empirical support for Proposition 1. Such patterns could be examined through systematic content analysis of CAPA reports, clinical incident investigations, or regulatory audit findings across regulated industries, drawing on methodological approaches used in algorithmic impact assessment research (Brown, Davidovic, & Hasan, 2021).

Proposition 2. Epistemic Authority

The stronger the perceived epistemic authority of computational outputs, the more rapidly responsibility expectations will align around computationally foregrounded interpretations.

Epistemic authority amplification occurs when algorithmically generated outputs are treated as objective, neutral, or technically superior forms of knowledge. Because computational representations derive legitimacy from formalized calculation and technical infrastructure, organizational actors may defer to these outputs with reduced contestation. Under conditions of causal ambiguity, the perceived precision of data-driven representations can accelerate interpretive alignment by stabilizing particular explanatory frames early in the sensemaking process. As actors converge around computationally foregrounded interpretations, alternative causal accounts may receive diminished scrutiny.

Empirically, this proposition could be examined by analyzing how computational outputs are treated within investigation documentation. Researchers could assess whether algorithmically generated records are cited without qualification, subjected to explicit contestation, or accompanied by interpretive caveats. Variation in attribution concentration across high- and low-authority computational environments, assessed through comparative content analysis of investigation reports, would provide empirical support for Proposition 2. Organizational contexts in which AI outputs carry formal procedural weight, such as systems integrated into regulatory filing templates or used as primary evidentiary inputs in CAPA documentation, represent high-authority conditions suitable for systematic comparison.

Proposition 3. Temporal Contraction

The shorter the temporal interval between event occurrence and formal computational documentation, the greater the likelihood of premature responsibility stabilization.

Temporal contraction of sensemaking refers to the reduction in the deliberative interval between an event and its formal inscription within documentation systems. Real-time dashboards, automated incident logging, and immediate classification tools reduce the period during which interpretive divergence may circulate informally among organizational actors. When documentation is generated rapidly and embedded within evaluative infrastructures, preliminary interpretations may acquire durability and institutional visibility. Responsibility accounts may therefore solidify before distributed or systemic explanations are meaningfully articulated.

Empirically, this proposition could be evaluated by comparing investigation outcomes across organizational settings that differ in the interval between event occurrence and formal documentation. Real-time logging environments represent conditions of high temporal contraction,

whereas settings relying primarily on retrospective narrative reporting represent lower contraction conditions. Convergence prematurity could be assessed by coding the proportion of investigation reports that rely primarily on computationally generated records as evidentiary anchors, alongside indicators of whether systemic or contextual explanations were examined prior to the formal attribution of individual responsibility.

Taken together, these propositions specify PAC as a conditional intensification dynamic within the responsibility lifecycle. Computational mediation does not generate responsibility convergence independently of existing organizational processes. Instead, it may accelerate stabilization and compress interpretive diversity under particular infrastructural conditions. Future empirical research could examine investigation archives, CAPA reports, or incident documentation across industries to assess whether PAC dynamics systematically correlate with computational documentation density and documentation timing. Such work would enable systematic evaluation of the structural conditions under which computational mediation accelerates responsibility convergence in organizational accountability systems.

Discussion: AI Mediation within the Responsibility Lifecycle

PAC introduces acceleration as an analytical dimension in sociotechnical and organizational theories of accountability. Prior research has examined how responsibility is anticipatorily aligned before action and how attribution stabilizes after events through investigative processes. Less attention has been given to how computational infrastructures reshape the tempo and breadth of stabilization across these stages. By conceptualizing AI as a representational accelerator rather than an autonomous decision-maker, PAC extends accountability theory beyond attribution and compliance toward the temporal dynamics through which responsibility is produced.

The central tension highlighted by PAC is between speed and deliberation. Computational systems increase administrative efficiency, yet they may simultaneously reduce the durability of ambiguity within organizational processes. Accountability systems differ not only in whom they assign responsibility to but also in how long interpretive variance is permitted to endure before stabilization becomes institutionally binding. When interpretive space contracts too rapidly, distributed or structural explanations may lose institutional durability before they are meaningfully examined.

This interpretation aligns with research on algorithmic governance showing that

computational systems frequently restructure decision environments and institutional reasoning processes rather than replacing human judgment outright (Pasquale, 2015; Kitchin, 2017; Raji et al., 2020). The central implication is not technological determinism but infrastructural acceleration. Responsibility convergence remains contingent on organizational ambiguity, closure pressure, and accountability design (Bovens, 2007). Computational mediation may intensify these dynamics when representational outputs are granted authority and rapidly embedded within documentation systems.

PAC is analytically complementary to Elish's (2019) moral crumple zone framework. The crumple zone concept identifies who absorbs responsibility when agency is distributed across automated systems. PAC instead examines when and how narrowly responsibility stabilization occurs across investigative processes. An organization may generate crumple zones without exhibiting PAC dynamics if responsibility is attributed slowly after extended deliberation. Conversely, PAC dynamics may occur without crumple zone effects if rapid stabilization targets systemic rather than individual causes. The two frameworks therefore identify distinct dimensions of AI-mediated accountability and point toward different forms of intervention.

This perspective also complements recent scholarship on accountability infrastructures (Raji et al., 2020; Brown, Davidovic, & Hasan, 2021), which demonstrates how auditing and documentation practices constitute accountability rather than merely recording it. PAC contributes by specifying how the temporal organization of documentation shapes the conditions under which responsibility stabilizes. From this perspective, AI systems function not only as decision-support tools but also as infrastructural components of organizational sensemaking environments. When computational systems compress both the representational range and the deliberative interval of post-event interpretation, they reshape accountability infrastructures in ways that may not be visible through audit frameworks focused primarily on bias, fairness, or transparency.

Viewing accountability as a temporal architecture of documentation, evidence, and interpretive durability highlights the importance of design choices within accountability systems. Preserving interpretive plurality may require mechanisms for contestability, temporal buffering, layered documentation that retains alternative explanations, and procedural conditions under which stabilized accounts can be revisited. Recognizing PAC as an infrastructural acceleration dynamic therefore enables more precise analysis of how accountability architectures evolve under computational mediation.

PAC ultimately foregrounds the temporal politics of responsibility stabilization. In AI-supported environments, responsibility may converge not because computational systems impose determinate judgments, but because they reshape the timing, authority, and durability of representational translation. Attending to this acceleration dynamic opens new avenues for designing accountability systems that balance administrative efficiency with sustained deliberative capacity.

Conclusion

This paper introduced Premature Accountability Convergence (PAC) as a conditional acceleration dynamic within contemporary accountability infrastructures. Rather than portraying AI as a determinant of responsibility, the framework conceptualizes computational systems as representational mediators that may compress interpretive space and intensify stabilization dynamics within the responsibility lifecycle. PAC therefore reframes the role of AI in organizations not as a source of new responsibility structures, but as an infrastructural force that reshapes the timing and scope through which responsibility converges.

Situating PAC within the broader responsibility lifecycle highlights that responsibility in regulated organizations does not emerge at a single moment of attribution. Instead, it is produced across interconnected stages through which expectations are aligned, evidence is structured, and causal accounts are stabilized. Computational mediation interacts with these stages by accelerating representational consolidation, particularly under conditions of ambiguity and epistemic authority amplification.

This perspective shifts the analytical focus of algorithmic accountability research. Rather than asking only whether computational systems generate correct or fair responsibility assignments, or whether they displace responsibility onto human operators, PAC directs attention to the temporal organization of accountability processes. The central question becomes how organizational infrastructures structure the timing of interpretation and stabilization, and how long interpretive plurality remains viable before institutional closure occurs.

Future research can build on this framework by examining investigation archives, CAPA reports, and incident documentation across industries to assess how documentation infrastructures shape responsibility stabilization. Comparative analysis of environments with different levels of computational documentation density may help clarify the conditions under which PAC

mechanisms emerge or are attenuated.

By foregrounding the temporal dimension of accountability infrastructures, PAC provides a conceptual foundation for analyzing how computational systems reshape the production and stabilization of responsibility in contemporary sociotechnical organizations.

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